

Denoising Permanently Shadowed Lunar Images with CNNs

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1 Introduction

The open project required denoising 19500 paired high-noise lunar images and producing predictions for 1000 noisy blind-test images. The main difficulty compared with the previous task was that the images are larger (128×128), the noise is stronger, and the clean test targets are not available. We therefore treated the task as supervised image-to-image regression on normalized grayscale intensities in $[0, 1]$, optimized validation MSE/PSNR on held-out training data, and inspected blind-test outputs visually.

Our final submission model is `model_01_unet_deep_01` from `results/final`. It is a large four-level U-Net and reached the best final validation score among the completed final reruns: MSE = 0.001758, MAE = 0.03213, and PSNR = 27.55 dB. The very similar `model_02` was also strong, but its final metrics were marginally lower.

2 Methods

Data handling and metrics. The paired train arrays and blind noisy validation array were loaded as tensors, converted to `float32`, divided by 255 when necessary, and given a channel dimension. Final reruns used a deterministic 5% validation split. Training was run on Euler GPUs with mini-batches and pinned-memory data loaders. On the held-out paired validation set, PSNR was computed in the standard full-reference way from the MSE between denoised output $\hat{x}$ and clean target x : $\text{PSNR} = 10 \log_{10}(1/\text{MSE}(\hat{x}, x))$, since images are normalized to $[0, 1]$. For the blind test set the clean image is unavailable, so this score cannot be interpreted as a true reconstruction PSNR. We still used a *proxy PSNR* against the noisy input, $\text{MSE}(\hat{x}, y_{\text{noisy}})$, as an indicative stability metric: it flags outputs that change the observation too little or too aggressively, while the real model ranking remains based on paired validation PSNR and visual inspection.

Final architecture. The selected model is a four-stage U-Net with feature widths $[64, 128, 256, 512]$ and a 1024-channel bottleneck. Each encoder/decoder stage uses two 3×3 convolutions with batch normalization and ReLU; max-pooling provides scale separation, transposed convolutions upsample, and skip connections concatenate encoder features into the decoder. This was well matched to denoising because the bottleneck captures broad noise statistics while the skips preserve crater rims and local albedo texture. The final layer is a 1×1 convolution followed by a sigmoid, keeping outputs in the normalized image range. The model has 31.0M trainable parameters. `/home/mattia/uni/space_data/space_data3/TASK_FINALcolored.pdf/home/`

Hyperparameter search with Optuna. After manual ablations, Optuna was used to search the most promising families. Each trial sampled a configuration, trained it for a shorter budget, evaluated validation PSNR, and reported intermediate scores to a median pruner, which stopped clearly uncompetitive trials early. For the Deep U-Net search, Optuna varied feature scale, dropout, output activation, learning rate, weight decay, optimizer, and batch size while keeping the loss fixed to MSE so PSNR comparisons remained consistent. The best Deep U-Net trials clustered around the large feature setting, AdamW, nonzero dropout/weight decay, and sigmoid output; the top Optuna PSNR for the selected final model was 27.6876 dB before the 150-epoch rerun.

3 Results

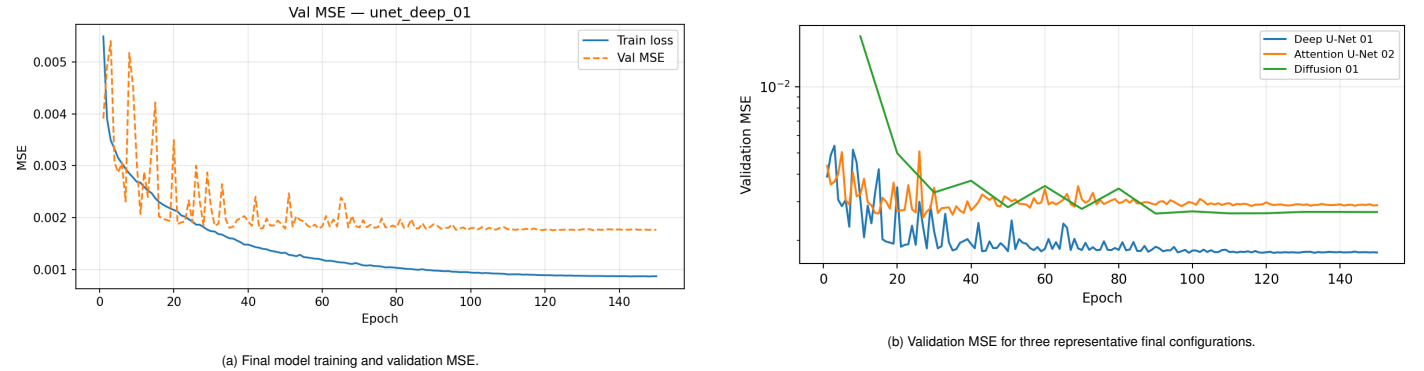


Figure 1: Training was stable and the final Deep U-Net separated clearly from the attention U-Net and diffusion reruns.

Table 2: Representative configurations. Lower MSE/MAE and higher PSNR are better.

Configuration	Main change	Best MSE	MAE	PSNR
Deep U-Net 01	large four-level U-Net, MSE, AdamW, dropout 0.109	0.001758	0.03213	27.55
Deep U-Net 02	same family, higher LR and dropout	0.001767	0.03223	27.53
Attention U-Net 02	attention-gated skips, combined MSE/L1	0.002542	0.03874	25.95
Diffusion 01	conditional DDPM-style denoiser, MSE	0.002654	0.03989	25.76

The broader ablation campaign also included baseline U-Nets, small/large channel counts, tanh vs. sigmoid outputs, Adam vs. AdamW, L1, MSE, mixed MSE/L1, and an MS-SSIM/L1 inspired loss. The strongest trend was architectural: multi-scale U-Nets with direct skip connections consistently outperformed flat ResNets and autoencoders because they remove global noise while reconstructing fine spatial structure. Attention gates were technically interesting but did not improve the final score here; they likely suppressed useful weak lunar texture along with noise. Diffusion models were the riskiest advanced attempt and produced plausible images, but validation was slow and their final MSE remained below the feed-forward Deep U-Net.

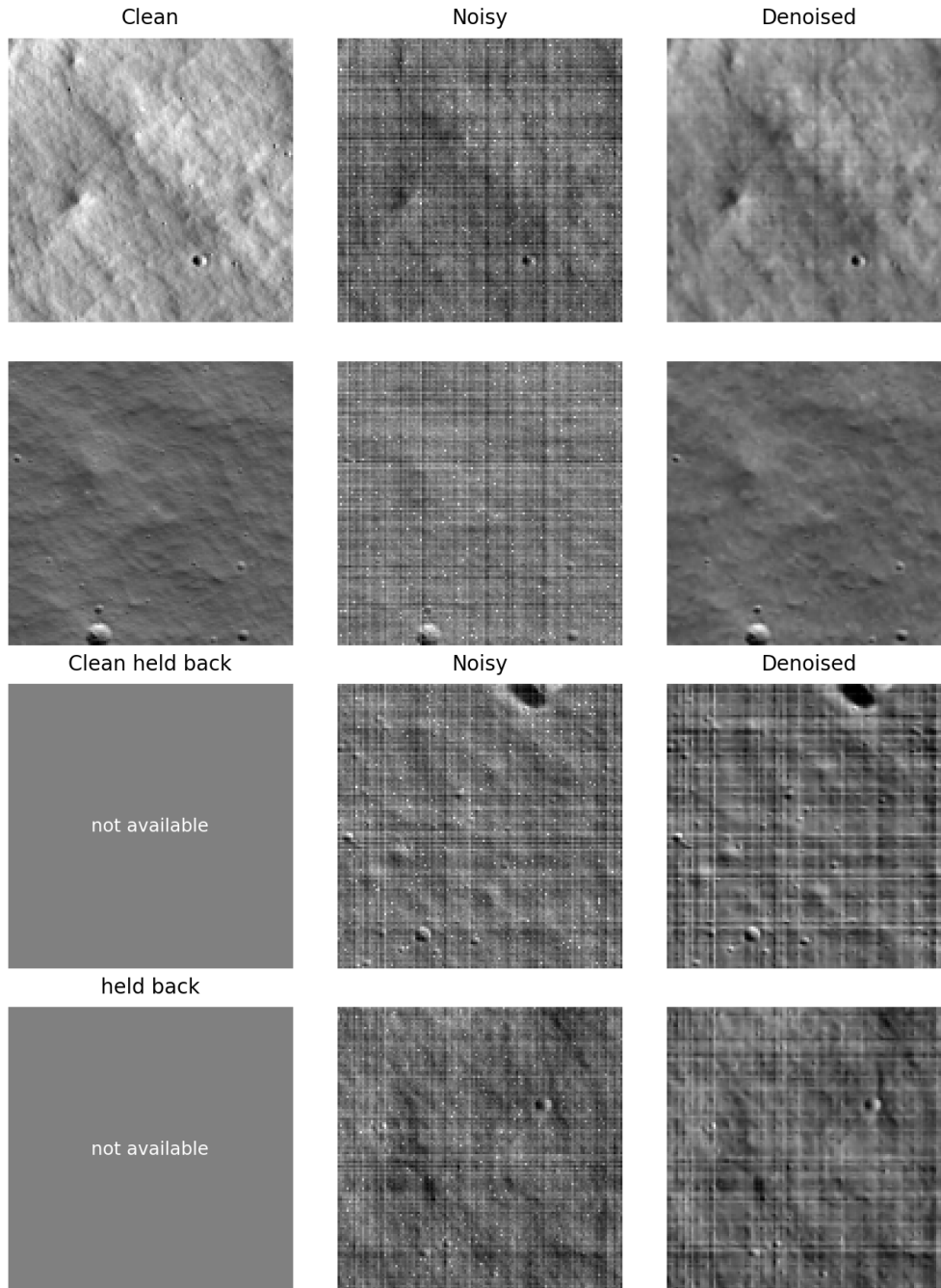


Figure 2: Two paired training examples and two blind-test examples using the final Deep U-Net checkpoint. Test clean images are held back by the assignment, so only noisy and denoised outputs can be shown there. The denoised images remove most high-frequency noise while preserving the dominant crater structures; some very fine contrast is still smoothed.

4 Discussion

The final model works best because it balances capacity and locality. The 31M-parameter Deep U-Net is large enough for the harder noise distribution, while convolutional structure and skip connections preserve geometry. MSE was selected because it directly optimizes the reported pixel-wise error and PSNR; L1 and combined losses sometimes looked sharper but did not beat MSE on validation metrics. AdamW and dropout helped control overfitting as validation loss plateaued.

The main limitation is validation protocol. The project statement asks for k-fold cross-validation, but the final artifacts available for this report contain deterministic hold-out reruns rather than full k-fold retraining of the selected 31M-parameter model. To reduce split sensitivity, we compared many architectures and repeated the strongest Deep U-Net region with closely related Optuna-selected configurations; `model_01` and `model_02` converged to almost identical scores.

5 Conclusion

We submit `model_01_unet_deep_01`, a large four-level U-Net trained with MSE and AdamW. It achieved the best completed final validation score in `results/final` and gave the cleanest trade-off between quantitative error, stable training, and visual preservation of lunar surface structure. The most useful exploration was not the exotic model alone, but the sequence from ablations to Optuna: manual experiments identified Deep U-Nets as the right family, and Optuna refined the learning rate, regularization, activation, and batch size inside that family.

Use of Generative AI

Generative AI tools were used for LaTeX editing, wording, and code/report assistance. The final content and reported metrics were checked against the local project files by the authors.